

1. (3.5 points)

Given the following information:

<u>Cumulative Loss Payments</u>			
Accident	12	24	36
<u>Year</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2013	1,500	2,700	3,450
2014	1,600	2,740	
2015	1,700		

- Exposures and premium are constant across all accident years.
- There is no development beyond 36 months.

a. (2 points)

Calculate the total reserve indication as of December 31, 2015 using loss-ratio based payout factors and the Benktander method.

b. (0.75 point)

Calculate the fifth-iteration Benktander method reserve indication for accident year 2015.

c. (0.75 point)

Assuming $Var[U_i] = Var[U_i^{BC}]$, use Hürlimann's method for optimal credibility and minimum variance to calculate the reserve indication for accident year 2015.

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QUESTION 1	
TOTAL POINT VALUE: 3.5	LEARNING OBJECTIVE: A1: Calculate unpaid claim estimates using credibility models.
SAMPLE ANSWERS	
Part a: 2 points	
<i>Sample Answer 1</i> Assume premium = 5,000 for each accident year. Other premium amounts may be assumed $M_1 = [(1,500 + 1,600 + 1,700)/3 \times 5,000] = 0.32$ $M_2 = [(1,200 + 1,140)/2 \times 5,000] = 0.234$ $M_3 = 750/5,000 = .15$ Expected loss ratio = 0.704. $U^0 = .704 \times 5,000 = 3,520$ $P_1 = .32/.704 = .455; Q_1 = 1 - .455 = .545$ $P_2 = (.32 + .234)/.704 = .787; Q_1 = 1 - .787 = .213$ $2014_{ind} = 2,740/.787 \times .213 = 742$ $2015_{ind} = 1,700/.455 \times .545 = 2,036$ $2014_{coll} = 3,520 \times .213 = 750$ $2015_{coll} = 3,520 \times .545 = 1,918$ $2014_{bt} = 742 \times .787 + 750 \times .213 = 743$ $2015_{bt} = 2,036 \times .455 + 1,918 \times .545 = 1,972$ Total reserve = $743 + 1,972 = 2,715$ <i>Sample Answer 2:</i> Avg. Paid: $0-12 \text{ months} = (1,500 + 1,600 + 1,700)/3 = 1,600$ $12-24 \text{ months} = (1,200 + 1,140)/2 = 1,170$ $24-36 \text{ months} = 750$ $U^0 = 1,600 + 1,170 + 750 = 3,520$ $P_1 = 1,600/3,520 = .455; Q_1 = 1 - .455 = .545$ $P_2 = (1,600 + 1,170)/3,520 = .787; Q_1 = 1 - .787 = .213$ 1st iteration ultimate losses $2014 = 3,520 \times .213 = 750; 750 + 2740 = 3,490$ $2015 = 3,520 \times .545 = 1,918; 1918 + 1,700 = 3,618$ 2nd iteration ultimate losses $2014 = 3,490 \times .213 = 743; 743 + 2740 = 3,483$	

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$$2015 = 3,618 \times .545 = 1,972; 1972 + 1,700 = 3,672$$

Total estimated Benktander outstanding losses as of December 31, 2015 =

$$3,483 + 3,672 - 2740 - 1700 = 2,715$$

Part b: 0.75 point

Sample Answer 1

$$\text{Expected Cost Reserves for AY 2015 } 2015_{EC} = (5,000) \times 70.4\% - 1,700 = 1,820$$

$$\begin{aligned} \text{Fifth Iteration Benktander Reserve} &= 2015_{ind} \times (1 - q^5) + 2015_{EC} \times q^5 \\ &= 2,036 \times (1 - 0.545^5) + 1,820 \times 0.545^5 \\ &= 2,025.6 \end{aligned}$$

Sample Answer 2

2nd iteration ultimate losses from part A

$$2015 = 3,618 \times 0.545 = 1,918; 1,918 + 1,700 = 3,672$$

3rd iteration ultimate losses from part A

$$2015 = 3,672 \times 0.545 = 2,001; 2,001 + 1,700 = 3,701$$

4th iteration ultimate losses from part A

$$2015 = 3,701 \times 0.545 = 2,017; 2,017 + 1,700 = 3,717$$

5th iteration ultimate losses from part A

$$2015 = 3,717 \times 0.545 = 2,026; 2,026 + 1,700 = 3,726$$

$$\text{Reserve} = 5^{\text{th}} \text{ iteration Ultimate minus paid} = 3,726 - 1700 = 2,726$$

Part c: 0.75 point

Sample Answer 1

$$Z = P_1 / (P_1 + \sqrt{P_1}) = 0.455 / (0.455 + \sqrt{0.455}) = 0.403$$

$$\text{Reserve} = Z \times 2015_{ind} + (1 - Z) \times 2015_{coll} = 0.403 \times 2,036 + 0.597 \times 1,918 = 1,966$$

Sample Answer 2

$$Z = P_1 / (P_1 + \sqrt{P_1}) = 0.455 / (0.455 + \sqrt{0.455}) = 0.403$$

$$\text{Estimated Ultimate} = 0.403 \times 3,736 + 0.597 \times 3,618 = 3,666$$

$$\text{Reserve} = \text{Estimated Ultimate minus Paid} = 3,666 - 1,700 = 1,966$$

EXAMINER'S REPORT

Candidates were expected to know how to produce loss ratio based payout factors and apply these factors in conjunction with the Benktander method. Many candidates lost credit for being unable to produce loss ratio based payout factors as outlined in Hurlimann. In addition, many candidates struggled to produce the correct *a priori* estimate.

Many candidates also lost credit for assuming the Benktander method was the first iteration of the calculation rather than the second.

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These items notwithstanding, most candidates performed reasonably well overall.

Part a

Candidates were expected to know how to produce loss ratio based payout factors and apply those factors in conjunction with the Benktander method. Many candidates struggled to produce loss ratio based payout factors and instead used a weighted average. Candidates also struggled to produce the appropriate *a priori* loss estimate. Many assumed a value of 3,450 since the oldest year was fully developed. However, this fails to recognize that payments for more recent years are emerging higher than corresponding payments for 2013. Once the payout factors and *a priori* estimate were derived, most candidates were able to compute the outstanding losses using the Benktander method.

Part b

Many candidates failed to understand that the Benktander method produces the second iteration reserve – not the first. Because of this, many candidates calculated the sixth iteration reserve and received partial credit.

Part c

The majority of candidates performed very well, receiving full credit.